Beyond the Machine: Building the Infrastructure That Connects Robotics

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Abstract

The robotics industry is advancing rapidly in mechanical design, artificial intelligence, autonomous control, sensing, and manufacturing. Yet the information infrastructure connecting these capabilities remains fragmented. Robotic components, assemblies, machines, software-defined capabilities, engineering configurations, and autonomous agents are frequently represented in separate systems, with inconsistent identifiers and incomplete continuity across procurement, deployment, maintenance, modification, and end-of-life recovery. This fragmentation creates a structural challenge: the industry needs more than increasingly capable machines; it needs persistent infrastructure for identifying, describing, relating, verifying, and managing the entities that constitute robotic systems throughout their lifecycles.

This paper introduces POWER IN THE NUMBERS (PITN), an overarching infrastructure framework designed to address this challenge through persistent identity, structured relationships, component intelligence, engineering context, and lifecycle continuity. PITN coordinates six complementary initiatives: ROBOPARTS, ROBOTRAITS, ROBOTRONIC, ROBOASI, ROBOAGENTIC, and ROBORECYCLE. Together, these initiatives address physical component identity and sourcing; robotic characteristics and capabilities; engineering integration and verification; artificial intelligence and knowledge infrastructure; autonomous-agent identity and orchestration; and industrial recovery, reuse, and circular lifecycle management.

The paper presents a five-layer architectural model and examines its relationship to established approaches including digital product passports, digital twins, decentralized identifiers, verifiable credentials, industrial automation standards, and robotics middleware. It also distinguishes the proposed architecture from the specific implementation mechanisms through which individual capabilities may be delivered. Publicly accessible project demonstrations and author-developed technical specifications provide evidence of architectural development, while the paper identifies

the additional testing, interoperability assessment, security evaluation, and independent validation required before broad production-readiness claims can be made.

The central contribution is a robotics-oriented architectural proposition: persistent identity is the foundation for maintaining meaningful relationships among components, machines, capabilities, engineering configurations, agents, and lifecycle records. The paper proposes a validation framework through which this architecture can be evaluated in representative robotics environments.

Keywords: robotics infrastructure; persistent identity; interoperability; digital product passport; component traceability; robotics lifecycle management; decentralized identifiers; verifiable credentials; autonomous agents; industrial data architecture; circular economy.

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1. Introduction

Robotics is entering a period of significant technological and industrial expansion. Humanoid robots, collaborative robots, autonomous mobile systems, industrial manipulators, intelligent end effectors, and increasingly capable robotic agents are developing alongside advances in artificial intelligence, machine vision, sensing, actuation, and manufacturing.

However, progress in the capabilities of individual robotic systems does not automatically create an interoperable industrial ecosystem. A robot is not an isolated machine. It is an assembly of components, subsystems, software, capabilities, engineering decisions, supplier relationships, operational records, and maintenance histories. These elements change over time and may be distributed across manufacturers, integrators, suppliers, operators, service organizations, and subsequent owners.

The information associated with these entities is frequently maintained in separate systems. Product catalogs, enterprise resource planning platforms, product lifecycle management systems, manufacturing execution systems, engineering repositories, maintenance applications, supplier databases, and machine-control environments each serve important purposes. Nevertheless, the relationships among their records are not necessarily persistent, consistent, or easily transferable.

A component may have a manufacturer part number, a supplier stock number, an internal engineering designation, and a separate maintenance identifier. A replacement component may be functionally similar without being technically equivalent. A robot assembly may be modified after deployment, while its original documentation remains unchanged. A capability description may be disconnected from the physical configuration that supports it. An autonomous agent may require an identifiable relationship to the system, equipment, permissions, and operating context within which it acts.

These conditions create a broader infrastructure problem. The challenge is not simply locating information. It is maintaining reliable relationships among information, physical assets, and the changing configurations of robotic systems.

POWER IN THE NUMBERS (PITN) was developed as an overarching framework to address this infrastructure challenge. Its central proposition is that persistent identity and structured relationships can provide a foundation for connecting otherwise fragmented robotics information.

The framework consists of PITN and six complementary initiatives:

1. ROBOPARTS — component identity, classification, sourcing, and lifecycle information.
2. ROBOTRAITS — robotic characteristics, capabilities, behavioral descriptions, and semantic representation.
3. ROBOTRONIC — engineering integration, configuration, verification, and performance context.
4. ROBOASI — artificial intelligence and structured knowledge infrastructure.
5. ROBOAGENTIC — autonomous-agent identity, orchestration, collaboration, and governance.
6. ROBORECYCLE — equipment recovery, component evaluation, reuse, and lifecycle recirculation.

PITN provides the overarching identity and relationship infrastructure connecting these initiatives.

The paper advances three research questions:

RQ1: How can persistent identity and structured relationships support interoperability across the physical, informational, engineering, and intelligent layers of robotics?

RQ2: How can component-level identity and lifecycle information be connected to robotic assemblies, capabilities, engineering configurations, and autonomous agents?

RQ3: What validation requirements are necessary to assess the technical feasibility, interoperability, reliability, and industrial applicability of a cross-layer robotics identity architecture?

This paper is an architectural perspective rather than a report of a completed empirical trial. Its contribution is the presentation of an integrated architectural proposition, the identification of relevant existing standards and technologies, and a proposed framework for subsequent technical validation.

2. The Interoperability Challenge in Robotics

2.1 Fragmented information across the robotics lifecycle

Robotic systems depend on information generated throughout their lifecycles. This information may include:

- Manufacturer and supplier identification.
- Component specifications and technical documentation.
- Bills of materials and assembly relationships.
- Engineering configurations and revisions.
- Software and firmware versions.
- Compatibility assessments.
- Installation and commissioning records.
- Maintenance and repair histories.
- Operational capabilities and limitations.
- Inspection, testing, and verification evidence.
- Ownership, transfer, reuse, and end-of-life records.

The existence of these records does not guarantee that they can be connected reliably.

Information may be duplicated across systems, manually reconciled, stored under inconsistent identifiers, or separated from the physical asset to which it relates. A record may describe a product family without identifying a particular physical instance. A supplier listing may provide a commercial description without establishing engineering compatibility. A maintenance record may refer to an internal equipment number that cannot be resolved outside the operator's organization.

These are not exclusively database problems. They involve identity, semantics, provenance, access, relationships, and governance.

2.2 Identity is not the same as interoperability

A persistent identifier can help establish continuity, but an identifier alone does not create interoperability.

For an identity architecture to support meaningful industrial exchange, it must be accompanied by structured descriptions, relationship models, data conventions, provenance information, and appropriate mechanisms for accessing or verifying associated records.

For example, identifying a robotic actuator is only an initial step. An industrial application may also need to determine:

- Which manufacturer produced it?
- Which model and revision does it represent?
- Which physical instance is being referenced?
- Which assembly contains it?
- Which controller and firmware configuration are associated with it?
- Which engineering requirements apply?

- What evidence supports its operating specifications?
- Has it been inspected, repaired, replaced, or modified?
- Is it approved for a particular application?

A useful architecture must preserve these relationships without assuming that every participating organization uses the same internal software or database.

2.3 The economic and operational significance

Information fragmentation can contribute to duplicated data entry, manual reconciliation, sourcing delays, engineering verification effort, incorrect component selection, maintenance inefficiencies, and incomplete asset histories.

The extent of these costs varies by organization and application. Their actual magnitude must be established through operational measurement rather than assumed from architectural design alone.

The proposed PITN framework addresses the underlying information structure that can contribute to these inefficiencies. Its economic value proposition should therefore be evaluated through measurable operational indicators, including information retrieval time, reconciliation effort, sourcing cycle time, configuration-verification effort, maintenance documentation completeness, and the cost of identifying suitable replacement components.

3. POWER IN THE NUMBERS: Overarching Infrastructure

PITN is the umbrella framework connecting six specialized initiatives. It is not intended to replace existing enterprise, engineering, manufacturing, or robotics systems. Rather, it proposes a persistent identity and relationship layer through which relevant information can be associated across those environments.

The architecture is based on three foundational concepts:

Persistent identity: A stable reference to an identifiable entity, whether a component, product model, physical machine, robotic system, capability, or autonomous agent.

Structured relationships: Explicit associations connecting entities to their components, suppliers, configurations, capabilities, evidence, ownership, and lifecycle events.

Lifecycle continuity: The ability to maintain relevant identity and relationship information as an entity is manufactured, assembled, deployed, serviced, modified, transferred, recovered, or retired.

The underlying architectural principle is:

The namespace is the entry point. Persistent identity is the infrastructure. The relationships are where the value accumulates.

This principle distinguishes a naming or discovery mechanism from the larger information architecture. A namespace may make an entity easier to identify or discover. Its industrial value depends on the quality of the records and relationships associated with that identity.

4. The Seven-Part Architecture

The framework comprises PITN and six specialized initiatives. Each addresses a different but connected domain.

4.1 PITN — Persistent identity and relationship infrastructure

PITN establishes the overarching identity architecture. Its proposed responsibilities include:

- Persistent identification and namespace organization.
- Identity resolution and discovery.
- Relationships among components, assemblies, machines, and agents.
- References to associated technical records and evidence.
- Cross-system information continuity.
- Lifecycle relationship management.
- Support for decentralized identity mechanisms where appropriate.

PITN is intended to provide a common architectural foundation without requiring all information to reside in one centralized repository.

4.2 ROBOPARTS — Physical component intelligence

ROBOPARTS addresses the physical components that make robotic systems possible.

Its architectural scope includes:

- Component classification and identification.
- Product definitions and physical-instance records.
- Manufacturer and supplier relationships.
- Bills of materials and assembly positions.
- Technical specifications and documentation references.
- Sourcing and request-for-quotation workflows.
- Compatibility assessment.
- Component lifecycle and maintenance records.
- Evidence references associated with product claims.

A critical distinction is made between a product definition and a physical instance. A product definition describes a model or type; a physical instance represents a particular manufactured or installed item.

This distinction is important for serial-level traceability, maintenance, repair, replacement, and configuration management.

ROBOPARTS is intended to support component discovery and information continuity across robotic systems. It does not imply that every listed component has been independently tested, certified, or verified for compatibility.

4.3 ROBOTRAITS — Robotic characteristics and capability representation

ROBOTRAITS addresses the structured description of robotic characteristics and capabilities.

Its proposed scope includes:

- Physical and functional characteristics.
- Capability descriptions.
- Behavioral traits and profiles.
- Operational contexts.
- Performance attributes and constraints.
- Confidence and provenance information.
- Semantic relationships among traits and capabilities.
- Versioned descriptions and validation references.

The purpose is to improve how robotic systems and agents are described and compared.

A trait or capability description should not be interpreted as evidence that a robot possesses a particular capability under all circumstances. Capability claims require defined operating conditions, configuration context, and supporting evidence.

The framework also distinguishes behavioral descriptions from claims about consciousness, subjective experience, or human-like understanding. Structured behavioral representation is an engineering and information-modeling concept, not a claim of human equivalence.

4.4 ROBOTRONIC — Engineering integration and verification

ROBOTRONIC addresses the engineering context connecting physical robotic systems, components, configurations, and performance requirements.

Its proposed scope includes:

- Engineering integration.
- System and subsystem configuration.
- Component compatibility.

- Configuration revision management.
- Performance metrics.
- Verification requirements.
- Testing evidence.
- Engineering change relationships.
- Relationships between simulation and physical testing.

A central consideration is that verification is configuration-dependent. Evidence obtained for one configuration should not automatically be assumed to apply after a material change to components, firmware, software, operating conditions, or system architecture.

Simulation, engineering analysis, and physical testing are distinct forms of evidence. The architecture must preserve that distinction.

4.5 ROBOASI — Artificial intelligence and knowledge infrastructure

ROBOASI addresses the artificial intelligence and structured knowledge dimension of the broader architecture.

Its proposed scope includes:

- Structured engineering knowledge.
- Declarative technical information.
- AI-assisted information processing.
- Knowledge relationships and contextual retrieval.
- Integration with robotics information domains.
- Support for distributed intelligence architectures.

The objective is to make relevant technical information more accessible and usable across the robotics ecosystem.

AI-generated outputs must remain distinguishable from authoritative source records, verified engineering data, and human-approved decisions. A system's ability to retrieve or generate information does not independently establish the accuracy of that information.

4.6 ROBOAGENTIC — Autonomous-agent identity and orchestration

ROBOAGENTIC addresses the growing role of autonomous software agents in robotics and industrial environments.

Its proposed scope includes:

- Agent identity and registration.
- Agent discovery.
- Mission and task representation.
- Delegation and collaboration.

- Authorization and permissions.
- Execution reporting.
- Outcome verification.
- Revocation and governance.
- Relationships between agents, machines, components, and operating environments.

An agent's identity must be distinguishable from its authority. Identifying an agent does not automatically authorize it to operate a robot, modify an engineering configuration, access protected information, or perform a physical action.

Authorization, safety controls, accountability, and execution verification require separate mechanisms.

4.7 ROBORECYCLE — Lifecycle recovery and circular industrial infrastructure

ROBORECYCLE addresses the recovery and recirculation of robotic equipment and components.

Its proposed workflow includes:

1. Identification of retired or surplus equipment.
2. Recovery and acquisition assessment.
3. Component identification.
4. Inspection and testing.
5. Condition classification.
6. Documentation of findings.
7. Qualification for reuse or refurbishment.
8. Inventory and disposition management.
9. Lifecycle record continuation.

This initiative connects the identity and information architecture to industrial reuse. Component identity and documented history can support evaluation, but they do not establish present condition or suitability. Those determinations require appropriate inspection, testing, and application-specific assessment.

5. Five-Layer Architectural Model

The seven-part framework can be organized into five functional layers.

Layer	Primary initiatives	Architectural responsibility

1. Persistent identity and relationships	PITN	Identity, namespace organization, resolution, provenance references, and cross-system relationships
2. Physical and semantic representation	ROBOPARTS, ROBOTRAITS	Component identity, product definitions, physical instances, characteristics, and capabilities
3. Engineering integration and verification	ROBOTRONIC	Configurations, engineering relationships, compatibility, testing, and performance evidence
4. Intelligence and governed autonomy	ROBOASI, ROBOAGENTIC	Knowledge processing, AI support, agent identity, orchestration, authorization, and governance
5. Lifecycle continuity and circularity	ROBORECYCLE	Recovery, inspection, condition records, reuse, refurbishment, and disposition

The layers are functional rather than necessarily separate software deployments. Their implementation may be distributed across services, repositories, databases, applications, and external systems.

5.1 Identity and relationship flow

A representative information flow begins with the identification of a component or robotic system.

The identity reference resolves to an appropriate record or service. That record may provide a product definition, physical-instance reference, or system identity. Structured relationships then connect the entity to its manufacturer, supplier, assembly, configuration, capabilities, engineering evidence, and lifecycle history.

Where applicable, AI systems and autonomous agents may use authorized references to retrieve or act upon relevant information. Subsequent maintenance, modification, inspection, or recovery events can contribute additional lifecycle records.

The intended result is continuity across organizational and technical boundaries, not the creation of a single universal database.

5.2 Separation of identity, data, and evidence

The architecture distinguishes three concepts:

- **Identity:** What entity is being referenced?

- **Data:** What information is associated with that entity?
- **Evidence:** What supports the accuracy, origin, or status of a particular claim?

These concepts should not be conflated. A persistent identifier does not prove the accuracy of a record. A database entry does not independently verify a physical asset. A cryptographic signature may establish that a credential was issued by a particular key holder, but it does not automatically establish that every statement within the credential is factually correct.

A robust implementation must therefore preserve provenance, issuer information, verification status, timestamps, and the scope of each claim.

6. Persistent Identity, Namespaces, and Web3 Mechanisms

6.1 Namespace-based discovery

Namespaces can provide memorable and structured references to robotics-related entities, categories, services, and information domains.

The PITN framework includes a namespace-oriented approach intended to support discovery and identity organization across robotics.

The `.humanoids` namespace provides a concrete example of this approach. The public storefront at Humanoids.select presents a branded name-discovery and purchasing workflow using reseller functionality provided by Freename.

The storefront allows visitors to search for a desired name, review availability and pricing, and proceed through the provider's checkout process. The described workflow is a hosted reseller-page arrangement.

It is important to distinguish three separate elements:

1. The public PITN-branded discovery experience.
2. The hosted Freename reseller page and checkout functionality.
3. Freename's separately available Reseller API integration route.

The hosted reseller workflow should not be described as a PITN-embedded API. The existence of a separate provider API does not establish that it has been integrated into PITN. The storefront demonstrates a branded discovery and reseller pathway; it does not, by itself, demonstrate a complete robotic identity or physical authentication system.

6.2 Decentralized identifiers

Decentralized identifiers (DIDs) provide a standards-based model for identifying subjects through identifiers that can be resolved to DID documents under supported methods.

The W3C DID Core Recommendation provides a relevant foundation for considering decentralized identity in robotics (W3C, 2022).

Potential applications include identifiers for:

- Robotic systems.
- Physical components.
- Manufacturers and suppliers.
- Software or autonomous agents.
- Organizations and authorized service providers.

The choice of DID method, resolver, key management arrangement, governance model, and recovery process must be determined for the intended deployment. A namespace name should not automatically be assumed to be a W3C-compliant DID.

6.3 Verifiable credentials

Verifiable credentials can provide a structured way to represent claims issued by an identifiable issuer and presented for verification.

The W3C Verifiable Credentials Data Model v2.0 provides a relevant standard for representing such claims (W3C, 2025).

In a robotics context, credentials could potentially represent:

- Manufacturer-issued component specifications.
- Inspection or test results.
- Maintenance certifications.
- Supplier attestations.
- Configuration approvals.
- Training or authorization records.

The trustworthiness of a credential depends on the issuer, the issuance process, the underlying evidence, and the verifier's policy. Cryptographic verification does not independently establish the factual truth of a claim.

6.4 Blockchain and ownership mechanisms

Blockchain-based naming, tokenization, and other Web3 mechanisms may provide useful options for identity registration, transfer, or decentralized reference management.

However, blockchain registration should not be equated with physical authentication, engineering certification, or operational authorization.

A digital name can reference a robotic asset without proving that the physical machine presenting that name is the asset described. A token transfer can represent a change in digital control or ownership under a particular system's rules without automatically updating legal title, manufacturer records, safety permissions, or service obligations.

The PITN architecture therefore treats decentralized mechanisms as possible components of a broader identity and evidence system, not as substitutes for engineering verification or governance.

7. Relationship to Existing Standards and Technologies

PITN is proposed as a complementary architecture. It does not claim to replace established standards, middleware, or enterprise systems.

7.1 Digital product passports

Digital product passports provide a developing framework for making product-related information accessible across lifecycle stages. Recent research emphasizes the need for reference architectures, semantic interoperability, data governance, and validation (Trienens et al., 2026; Hbeich et al., 2026; Lyse & Huang, 2026).

The PITN architecture shares the objective of persistent product information but extends the discussion into robotics-specific relationships among physical components, assemblies, engineering configurations, capabilities, and autonomous agents.

A digital product passport may serve as an important representation or access mechanism within this architecture. It is not equivalent to the entire PITN framework.

7.2 Digital twins

ISO 23247-1:2021 establishes an overview and general principles for digital twin frameworks in manufacturing.

Digital twins address relationships between physical entities and their digital representations. PITN's proposed contribution is complementary: persistent identity and structured relationships can help maintain references to the physical entity, its components, configuration, evidence, and lifecycle records.

An identity record is not automatically a digital twin. A digital twin requires an appropriate digital representation and defined relationship to the physical system.

7.3 Industrial automation and enterprise integration

ISA-95, also known as IEC 62264, provides models and terminology for enterprise-control system integration.

PITN does not seek to replace manufacturing execution, enterprise resource planning, or control-system architectures. Its proposed role is to support persistent references and relationships across the information generated and maintained by these systems.

7.4 Robotics middleware

ROS 2 and ROS 2 Control provide established software and hardware-interface frameworks for robotics.

ROS 2 Control includes hardware abstractions and lifecycle mechanisms relevant to system and component integration. These technologies address operational software interfaces and control-related concerns. PITN addresses a different architectural concern: persistent identity and information relationships that may span beyond an individual middleware environment.

No claim of completed native integration between PITN and ROS 2 Control is made in this paper.

7.5 Semantic interoperability

Semantic interoperability requires more than exchanging data in a technically readable format. Participating systems must also interpret the exchanged information consistently.

Hbeich et al. (2026) identify semantic interoperability as a central challenge for digital product passports. Kranner and Al-Akrawi (2026) examine the relationship among data spaces, existing standards, and initiatives. Havur et al. (2026) investigate the integration of UN/CEFACT Transparency Protocol semantics into Asset Administration Shell submodels.

These developments are relevant to PITN because persistent identity must be accompanied by shared or mappable meanings for the information associated with each entity.

The architecture therefore requires future work on controlled vocabularies, ontology alignment, data schemas, mapping rules, and version management.

7.6 Related decentralized product-passport research

Babel et al. (2026) present a decentralized, Web3-inspired digital product passport architecture focused on long-term data access and consumer data sovereignty. Their design combines digital wallets, verifiable credentials, and non-fungible tokens to support issuer-independent data transferability and ownership management.

This work is directly relevant to the PITN discussion of decentralized identity and lifecycle continuity. It also illustrates that decentralized product-data architectures are an active area of research.

The principal emphasis differs. Babel et al. focus on consumer sovereignty and product-data custodianship across ownership transitions. PITN proposes a broader robotics-oriented architecture connecting component hierarchies, physical-instance identity, engineering configurations, capability representation, autonomous-agent governance, and industrial recovery.

The two approaches should be understood as related architectural work addressing overlapping but distinct problem scopes.

8. Proposed Data and Relationship Model

A persistent identity architecture requires a data model capable of representing different entity types and their relationships.

The proposed PITN model distinguishes at least the following entity classes:

Entity class	Example	Relevant relationships
Product definition	Actuator model	Manufacturer, specifications, revisions
Physical instance	Serialized actuator	Product definition, serial number, inspection history
Component position	Joint actuator position	Assembly, bill of materials, configuration
Robotic system	Humanoid robot	Components, subsystems, software, capabilities
Capability	Manipulation capability	Robot, configuration, operating conditions, evidence
Engineering configuration	Approved revision	Components, firmware, software, test records
Autonomous agent	Mission-planning agent	Identity, permissions, tasks, execution records
Lifecycle event	Repair or replacement	Entity, timestamp, actor, evidence
Evidence record	Inspection report	Issuer, subject, date, method, result

The model must distinguish between a product type and a physical instance. It must also preserve the difference between a relationship asserted by a system and a relationship independently verified through evidence.

8.1 Provenance and evidence references

Each significant claim should, where practical, be associated with:

- The identity of the issuing organization or individual.
- The subject of the claim.
- The date and relevant time context.
- The method or process used to establish the claim.
- The supporting document, dataset, or test record.
- The applicable configuration or revision.
- The status and scope of verification.

This supports traceability without requiring all underlying documents to be stored in a single system.

8.2 Versioning and change management

Persistent identity does not mean that every attribute remains unchanged.

Components may be revised, repaired, replaced, or reconfigured. Systems may receive software updates, new sensors, or different actuators. Capability descriptions may change as a result.

The architecture should therefore preserve stable entity references while recording versioned attributes, configuration changes, and lifecycle events.

A change to one component should not silently overwrite the historical record of the previous configuration.

9. Public Demonstrations and Author-Developed Technical Materials

The PITN framework is supported by author-developed technical documentation and publicly accessible project demonstrations. These materials are original project work authored by Aaron Bailot.

The following examples illustrate different aspects of the architecture:

9.1 PITN demonstration environment

Public demonstration: https://pitn.ai/#demo

This provides a public project demonstration entry point. Individual capabilities should be assessed separately, and the presence of a demonstration interface should not be interpreted as proof that every proposed subsystem is fully implemented or independently validated.

9.2 Humanoids.select

Public discovery experience: https://www.humanoids.select

Humanoids.select demonstrates a branded `.humanoids` name-discovery and reseller workflow using Freename's hosted reseller functionality.

The provider's reseller documentation describes a public storefront workflow involving domain discovery, cart functionality, and checkout. Freename also documents its Reseller API as a separate integration route. The hosted reseller page is not evidence of an embedded PITN API or of physical robot authentication.

9.3 Factory01 passport

Public passport example: https://powerinthenumbers.ai/passport/factory01

The Factory01 example illustrates structured identity and passport presentation for a humanoid robotics context.

A displayed identity or passport record is evidence of information presentation. It does not independently establish physical-device authentication, on-chain verification, safety certification, or operational authorization.

9.4 Identity scan

Public identity experience: https://powerinthenumbers.ai/identity#scan

The identity scan experience illustrates a proposed path from public discovery and identity reference to passport access through a QR-based interaction.

A QR code is a means of accessing or resolving information. It is not, by itself, a cryptographic trust mechanism or proof that the physical object is authentic.

9.5 Author-developed technical documentation

The architecture is further documented in author-developed technical specifications and project repositories, including:

- PITN namespace and identity architecture.
- ROBOPARTS data model and API/database prototype specification.
- ROBOTRAITS trait ontology and technical specifications.

- ROBOTRONIC engineering integration and verification specification.
- ROBOASI intelligence and knowledge architecture specifications.
- ROBOAGENTIC autonomous-agent architecture and governance specification.
- ROBORECYCLE operating framework and lifecycle recovery specifications.

These materials are project documentation authored by Aaron Bailot. They should be evaluated as technical specifications and development artifacts, not as independently peer-reviewed publications.

The presence of a specification establishes that a capability has been described architecturally. It does not, without further evidence, establish that the capability has been implemented, tested, or independently verified.

10. Implementation Maturity and Evidence Classification

A consistent maturity vocabulary is necessary to prevent architectural proposals, demonstrations, and verified implementations from being treated as equivalent.

This paper adopts the following classifications:

Classification	Meaning
Specified	A function or architecture is described in identifiable technical documentation.
Prototyped	An experimental artifact or preliminary implementation exists.
Implemented	The function exists in an executable or running implementation.
Tested	Defined tests have been executed and results documented.
Independently verified	An independent party has assessed the function against stated criteria.
Operationally demonstrated	The function has been observed in a real or controlled operating environment.
Commercially evidenced	Commercial activity is supported by transaction, agreement, invoice, or provider records.

These classifications are not interchangeable.

A prototype specification is not proof of a tested API. A public demonstration is not independent verification. A reseller checkout workflow is not proof of completed sales or revenue. A proposed integration is not evidence of an operational integration.

For the PITN framework, the public demonstrations described above provide evidence of identity presentation, discovery, and passport-access concepts. The author-developed specifications establish the documented architectural scope of the initiatives. Broader implementation and validation claims require reproducible technical evidence.

This distinction is particularly important for robotics, where errors in component compatibility, configuration assumptions, safety-related information, or agent authorization can have physical consequences.

11. Proposed Validation Framework

The principal next step is systematic validation across representative robotics use cases.

The following framework identifies the tests necessary to evaluate the architecture. These are proposed validation activities, not reported completed test results.

11.1 Identity persistence and resolution

Objective: Determine whether entity identifiers remain consistently resolvable across relevant lifecycle events.

Potential measures:

- Identifier uniqueness within the defined namespace or registry.
- Resolution success rate.
- Resolution latency.
- Duplicate and conflicting identity detection.
- Recovery behavior following service or provider changes.
- Continuity after ownership or organizational transfer.

11.2 Component and assembly traceability

Objective: Determine whether physical instances can be connected reliably to product definitions, assembly positions, and lifecycle records.

Potential tests:

- Product-definition versus physical-instance distinction.
- Serial-level identification.
- Bill-of-materials relationship integrity.
- Component replacement and removal.
- Historical configuration reconstruction.
- Lifecycle-event association.

11.3 Semantic interoperability

Objective: Determine whether different systems can interpret exchanged records consistently.

Potential tests:

- Schema mapping.
- Controlled-vocabulary alignment.
- Ontology compatibility.
- Missing or conflicting attribute handling.
- Version compatibility.
- Cross-platform data exchange.

11.4 Engineering configuration and evidence

Objective: Determine whether engineering evidence remains associated with the correct system configuration.

Potential tests:

- Configuration revision tracking.
- Component substitution.
- Firmware and software version changes.
- Verification evidence association.
- Revalidation triggers after changes.
- Distinction between simulation, inspection, and physical test evidence.

11.5 Autonomous-agent identity and governance

Objective: Determine whether agent identity, authorization, task assignment, and execution records can be represented and audited.

Potential tests:

- Agent registration and identity resolution.
- Permission enforcement.
- Delegation boundaries.
- Revocation behavior.
- Task-to-agent attribution.
- Execution reporting and evidence.
- Unauthorized action prevention.

11.6 Lifecycle recovery and reuse

Objective: Determine whether identity and lifecycle information can support equipment recovery and component reuse decisions.

Potential tests:

- Recovery record completeness.
- Inspection and condition documentation.
- Component history retrieval.
- Reuse eligibility workflow.
- Replacement-part traceability.
- Disposition and ownership records.

11.7 Security, privacy, and resilience

Objective: Evaluate whether the architecture protects information and identity services against relevant threats.

Potential tests:

- Authentication and authorization controls.
- Credential issuance and revocation.
- Key-management and recovery processes.
- Data integrity and provenance.
- Access-control enforcement.
- Privacy and selective disclosure.
- Service availability and recovery.
- Threat modeling and penetration testing.

11.8 Industrial pilot design

A practical pilot should begin with a bounded robotic system rather than attempting to validate every initiative simultaneously.

A representative pilot could include:

1. One robotic system with a defined configuration.
2. A selected group of serialized components.
3. Product definitions and supplier references.
4. A bill of materials.
5. A limited set of capability descriptions.
6. Engineering and inspection records.
7. A documented maintenance or component-replacement event.
8. A defined identity-resolution and evidence-access workflow.

The pilot should establish baseline performance and compare the proposed identity architecture with the organization's existing information processes.

Results should include test procedures, system versions, datasets, measured outcomes, known limitations, and reproducible evidence.

12. Research Propositions

The architecture supports several propositions suitable for future empirical examination.

P1 — Identity continuity: Persistent identifiers combined with structured lifecycle events can improve the continuity of component and system information across organizational boundaries.

P2 — Relationship completeness: Explicit relationships among product definitions, physical instances, assemblies, configurations, and evidence can improve the reconstruction of robotic system histories.

P3 — Semantic interoperability: Identity resolution combined with shared or mapped semantic models can improve the consistency of information exchange among heterogeneous robotics systems.

P4 — Configuration-aware verification: Linking verification evidence to specific configurations can improve the identification of changes that require renewed engineering assessment.

P5 — Governed autonomy: Explicit agent identity, authorization, delegation, and execution records can improve the auditability of autonomous-agent activity.

P6 — Lifecycle continuity: Persistent identity and structured condition records can support more informed recovery, refurbishment, and reuse decisions.

These propositions are hypotheses for testing. They are not presented as empirically established findings.

13. Limitations

Several limitations must be acknowledged.

First, the proposed architecture spans multiple technical domains. Its components may require different data models, governance arrangements, security mechanisms, and implementation environments. A unified conceptual framework does not guarantee seamless technical integration.

Second, persistent identity does not guarantee data accuracy. The architecture depends on reliable issuers, appropriate validation processes, and evidence quality.

Third, decentralized naming and identity mechanisms introduce dependencies involving resolver availability, key management, governance, provider continuity, and recovery.

Fourth, semantic interoperability requires more than identifier assignment. Shared meanings, schema mappings, controlled vocabularies, and version management remain necessary.

Fifth, physical robotics systems introduce safety and reliability requirements that cannot be satisfied through identity infrastructure alone. Engineering validation, functional safety, cybersecurity, and operational authorization must remain within their appropriate technical and regulatory frameworks.

Sixth, the public demonstrations and author-developed specifications discussed in this paper do not establish that all seven architectural elements are integrated into a single production system. Their maturity must be assessed at the level of individual functions and supported by reproducible evidence.

Seventh, the proposed operational and economic benefits have not been established through a controlled comparative study. Quantitative claims about cost reduction, productivity, or return on investment require baseline measurements and pilot results.

Finally, the architecture must be evaluated in the context of existing industrial systems. Adoption will depend on whether it provides measurable value without imposing unnecessary duplication, complexity, or administrative burden.

14. Future Research

Future research should prioritize the following areas:

1. **Persistent identity implementation:** Evaluate identifier schemes, resolution methods, and recovery mechanisms across heterogeneous robotics environments.
2. **Semantic model development:** Establish mappings among component taxonomies, product data models, robotics capabilities, and existing standards.
3. **Configuration-aware evidence:** Develop methods for associating test results, certifications, and engineering records with precise system configurations.
4. **Digital passport interoperability:** Evaluate how robotics-specific identity and lifecycle records can align with emerging digital product passport architectures.
5. **Autonomous-agent governance:** Investigate identity, authorization, revocation, accountability, and execution evidence for agents interacting with physical systems.
6. **Lifecycle and circularity pilots:** Measure whether persistent component histories improve recovery, inspection, refurbishment, and reuse workflows.
7. **Security and resilience:** Conduct threat modeling, independent security review, and long-term identity-service testing.
8. **Economic evaluation:** Measure information retrieval, reconciliation, sourcing, engineering verification, and maintenance costs before and after pilot deployment.

A successful validation program should include robotics manufacturers, component suppliers, system integrators, operators, maintenance providers, and relevant standards organizations.

15. Conclusion

The future of robotics depends not only on more capable machines but also on the infrastructure that connects their components, identities, capabilities, configurations, information, and lifecycle histories.

This paper has introduced POWER IN THE NUMBERS (PITN) as an overarching architecture for addressing that challenge. Through six complementary initiatives—ROBOPARTS, ROBOTRAITS, ROBOTRONIC, ROBOASI, ROBOAGENTIC, and ROBORECYCLE—the framework proposes a connected approach to physical component identity, semantic representation, engineering integration, artificial intelligence, autonomous-agent governance, and lifecycle recovery.

Its central architectural contribution is the proposition that persistent identity and structured relationships can provide continuity across otherwise fragmented robotics information environments.

The framework is intended to complement existing industrial standards, enterprise systems, digital product passport initiatives, digital twin approaches, decentralized identity technologies, and robotics middleware. It does not claim to replace them. Instead, it proposes a common identity and relationship foundation through which their information can be associated and interpreted across the lifecycle of robotic systems.

The distinction between identity, information, and evidence is fundamental. A name does not authenticate a physical machine; a record does not prove a component's condition; a credential does not automatically establish the truth of every claim; and an agent identifier does not confer operational authority. These distinctions must remain explicit in any implementation intended for industrial use.

The proposed architecture therefore represents both an infrastructure proposition and a research agenda. Its value must be established through reproducible testing, interoperability assessment, security review, engineering pilots, and measurable operational outcomes.

The long-term objective is a robotics ecosystem in which components, machines, capabilities, engineering configurations, intelligent agents, and lifecycle records can retain meaningful identity and relationships as they move across systems, organizations, and stages of industrial use.

The machine may perform the task. The infrastructure must preserve the identity, context, relationships, and evidence that make the task understandable, traceable, and accountable.

References

Peer-reviewed academic literature

Babel, M., Guthmann, C., Körner, M.-F., Kranz, T., & Strüker, J. (2026). Whose passport is it anyway? A decentralized web3-inspired architecture for sovereign digital product passports in the circular economy. *Electronic Markets, 36*, Article 84. https://doi.org/10.1007/s12525-026-00927-7

Havur, G., Pellegrini, T., Schenner, G., & Fusco, F. (2026). A design study on integrating UNTP semantics into AAS submodels for digital product passports. *Procedia Computer Science, 277*, 1130–1139. https://doi.org/10.1016/j.procs.2026.02.153

Hbeich, E., Bernier, C., Van Hooland, S., Robal, T., & Maigre, R. (2026). Achieving semantic interoperability for digital product passports. *Procedia Computer Science, 277*, 2981–2992. https://doi.org/10.1016/j.procs.2026.02.334

Kranner, M., & Al-Akrawi, A. (2026). Bridging the gaps towards a unified architecture for digital product passports: A data space perspective on existing standards and initiatives. In F. Longo, V. Solina, & E. Francalanza (Eds.), *Procedia Computer Science, 277*, 2374–2387. https://doi.org/10.1016/j.procs.2026.02.274

Lyse, S. M., & Huang, L. (2026). Digital product passports: A systematic literature review on framework design and validation. *Digital, 6*(2), Article 43. https://doi.org/10.3390/digital6020043

Trienens, M., Wyrwich, F., Schörder, L., Hovemann, A., & Dumitrescu, R. (2026). Towards a reference architecture for digital product passports in engineering. *Proceedings of the Design Society, 6*, 1601–1610. https://doi.org/10.1017/pds.2026.10518

Valtanen, K., Ylikerälä, M., & Hakola, L. (2026). *Full-stack digital product passport architecture supporting interoperability and AI-enabled use* (Version 1). Zenodo. https://doi.org/10.5281/zenodo.20848510

Standards and technical specifications

International Organization for Standardization. (2021). *ISO 23247-1:2021: Automation systems and integration—Digital twin framework for manufacturing—Part 1: Overview and general principles.* https://www.iso.org/standard/75066.html

International Society of Automation. (2025). *ANSI/ISA-95.00.01-2025: Enterprise-control system integration—Part 1: Models and terminology.* ISA.

World Wide Web Consortium. (2022). *Decentralized identifiers (DIDs) v1.0: Core architecture, data model, and representations.* W3C Recommendation. https://www.w3.org/TR/did-core/

World Wide Web Consortium. (2025). *Verifiable credentials data model v2.0.* W3C Recommendation. https://www.w3.org/TR/vc-data-model-2.0/

Open Robotics. (n.d.). *ROS 2 Control documentation.* https://control.ros.org/

Freename AG. (2026, February 2). *How the Freename reseller page works.* https://freename.com/help/how-the-freename-reseller-page-works

Author-developed PITN technical documentation

The following materials are original project documentation authored by Aaron Bailot. They are cited as technical specifications, project documentation, or unpublished author-developed materials—not as peer-reviewed publications.

Bailot, A. (n.d.). *PITN.ai—RoboParts.ai: Persistent digital identity and lifecycle infrastructure for robotics* [Project overview and repository documentation]. POWER IN THE NUMBERS. https://github.com/PITN374/powerinthenumbers

Bailot, A. (n.d.). *PITN—27-namespace humanoid robot architecture* [Technical architecture specification]. POWER IN THE NUMBERS. https://github.com/PITN374/ROBOPARTS.ai

Bailot, A. (n.d.). *ROBOPARTS data model* (Version 0.1.0) [Technical data-model specification]. ROBOPARTS. https://github.com/PITN374/ROBOPARTS.ai/blob/main/ROBOPARTS_DATA_MODEL.txt

Bailot, A. (n.d.). *ROBOPARTS.ai—API and database prototype* (Document RP-API-010, Version 1.0) [Prototype specification]. ROBOPARTS. https://github.com/PITN374/ROBOPARTS.ai/blob/main/API_AND_DATABASE_PROTOTYPE.txt

Bailot, A. (2026, October 11). *IP-011—ROBOTRAITS trait-based robotic intelligence ontology* (Proposed ontology release 3.1.0) [Technical ontology specification]. ROBOTRAITS.

Bailot, A. (n.d.). *ROBOTRONIC engineering integration, configuration verification, and performance metrics specification* (Document RB-ENG-002, Version 1.0.0) [Proposed engineering specification]. ROBOTRONIC. Private project documentation.

Bailot, A. (n.d.). *ROBOASI robotics intelligence: Consolidated declarative engineering intelligence and intellectual property specification* (Version 1.0) [Proposed architectural specification]. ROBOASI. Private project documentation.

Bailot, A. (n.d.). *ROBOAGENTIC autonomous agent orchestration, collaboration, and governance infrastructure: Foundational intellectual property and technical architecture specification* [Proposed architecture and functional specification]. ROBOAGENTIC. Private project documentation.

Bailot, A. (n.d.). *ROBORECYCLE.ai: Definitive operating framework and robotics manufacturer benefits—Industrial technology recovery, validation, reuse, and recirculation* [Operating framework]. ROBORECYCLE. https://github.com/PITN374/ROBORECYCLE/blob/main/RRframework.txt

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Author Statement

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Originating framework: POWER IN THE NUMBERS (PITN)

The author is the originator of the PITN framework and the six associated initiatives discussed in this paper: ROBOPARTS, ROBOTRAITS, ROBOTRONIC, ROBOASI, ROBOAGENTIC, and ROBORECYCLE. The project architectures, concepts, specifications, and associated technical materials described herein are author-developed work.

Author contribution: Conceptualization, architectural design, development of the framework, technical documentation, project demonstrations, and preparation of the manuscript.

Competing interests: The author is the originator of the PITN framework and its associated initiatives and has a direct interest in their development and potential commercialization. This relationship is disclosed for transparency.

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Submission Note

This manuscript is prepared as an original technical architectural perspective with a proposed validation framework. It distinguishes author-developed architectural contributions, publicly accessible demonstrations, external scholarly literature, and future validation requirements. It does not present proposed capabilities as independently verified empirical results.

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